AP PHYSICS 1

Unit 1 • Lesson 1.2 • Guided notes

Topic 1.2 • Displacement, Velocity, and Acceleration

Guided notes • Fill in the blanks and work the examples in the boxes

Objective: define velocity and acceleration, and use the kinematic equations for constant acceleration.

Definitions	Average velocity: $v_{\text{avg}} =$ _____ Average speed = _____ Average acceleration: $a_{\text{avg}} =$ _____ Units: v in _____, a in _____ Acceleration is the rate of change of _____: how much the velocity changes _____.
Speeding up or slowing down?	v and a have the **same** sign: the object is _____. v and a have **opposite** signs: the object is _____. A negative acceleration does _____ always mean slowing down: a car moving in the $-x$ direction with $a < 0$ _____.
The kinematic equations (constant a only)	1) $v =$ _____ no Δx 2) $\Delta x =$ _____ no v 3) $v^2 =$ _____ no t Strategy: list the known quantities (v_0, v, a, t, Δx), find the one that is _____, pick the equation without it.
Example 1: a car speeds up	A car goes from 4 m/s to 16 m/s in 3 s with constant acceleration. $a = \dfrac{\Delta v}{\Delta t} =$ _____ $\Delta x = v_0 t + \frac{1}{2}at^2 =$ _____
Example 2: braking	A car moving at 20 m/s brakes with $a = -5\,\text{m/s}^2$ until it stops. Time not given: use equation ____. $0 = 20^2 + 2(-5)\Delta x \Rightarrow \Delta x =$ _____ Time to stop: $t =$ _____
Free fall	Near Earth, an object in free fall (no air resistance) has $a =$ _____ _____, whether it moves up or down. A stone is dropped from rest. After 2 s: $v =$ _____ downward; it has fallen _____.
Summary	